

Wastewater INfluent Dynamics (WIND) Model: A Modular Framework for Simulating Influent Flow Dynamics in Urban Sewer System

by

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This document describes how to run WIND model for influent generation.

To run the reservoir method

- Open [WIND_model \(reservoir_method\).ipynb](#)
- Upload CSV file with 'Date', 'Rainfall (mm/day)', and 'seasonal' columns in input folder. If your facility does not show seasonality, you put 1 to the all rows of seasonal column
- Set the working directory and change the csv file name
- Set weekly fraction for domestic and industrial wastewater
- Set population and Daily per capita wastewater production
- Set average daily industrial wastewater
- Set K_imp for direct runoff
- Set a_loss, a_inf, h_inf_init, and K_inf for infiltration
- Set alpha if you have seasonality
- Set Kr

To run the reservoir method

- Open [WIND_model \(exponential_delay_method\).ipynb](#)
- Upload CSV file with 'Date', 'Rainfall (mm/day)', and 'seasonal' columns in input folder. If your facility does not show seasonality, you put 1 to the all rows of seasonal column
- Set the working directory and change the csv file name
- Set weekly fraction for domestic and industrial wastewater
- Set population and Daily per capita wastewater production
- Set average daily industrial wastewater
- Set K_imp for direct runoff
- Set kd, and K_inf for infiltration
- Set alpha if you have seasonality
- Set Kr

Note: The script is thoroughly commented to improve clarity and make it easier for users to understand and customize each modeling step.